Automatic Night Lamp Controller PCB

Designed using KiCad

Name of the Student:	Arnav Tandon
Title:	Automatic Night Lamp Controller PCB
Tool:	KiCad
Organisation:	LNT EduTech

1. Project Overview and Working Principle

This project is an automatic night lamp that switches an LED on in low light and off in daylight. A light dependent resistor (LDR) and a fixed resistor (R1) form a voltage divider that drives the base of an NPN transistor (Q1, BC547).

In daylight: the LDR resistance is low, so the base node is pulled close to ground. The transistor stays off and the LED is off.

In darkness: the LDR resistance is high, so base current flows through R1. The transistor turns on and the LED lights, with R2 limiting the LED current.

Design calculation (5 V supply)

Base current in the dark = (5 - 0.7) / 47k = approx. 90 uA. With hFE of at least 110 this is enough to saturate Q1 for an LED current of about 10 mA.
R2 = (5 - 2) / 10 mA = 300 ohm, so the standard value 330 ohm is used.
The lamp switches on when the LDR resistance rises above roughly 7 kohm (dim light).

Circuit connections

Ref	Connection
J1	Pin 1 to +5V, pin 2 to GND
R1 (47k)	Between +5V and base node
LDR1	Between base node and GND
Q1 (BC547)	Base to base node, emitter to GND, collector to LED cathode
D1 (LED)	Cathode to Q1 collector, anode to R2
R2 (330R)	Between LED anode and +5V

2. Schematic

Symbols used: Connector:Conn_01x02_Pin, Device:R, Device:R_Photo, Transistor_BJT:BC547, Device:LED, +5V, GND, PWR_FLAG.

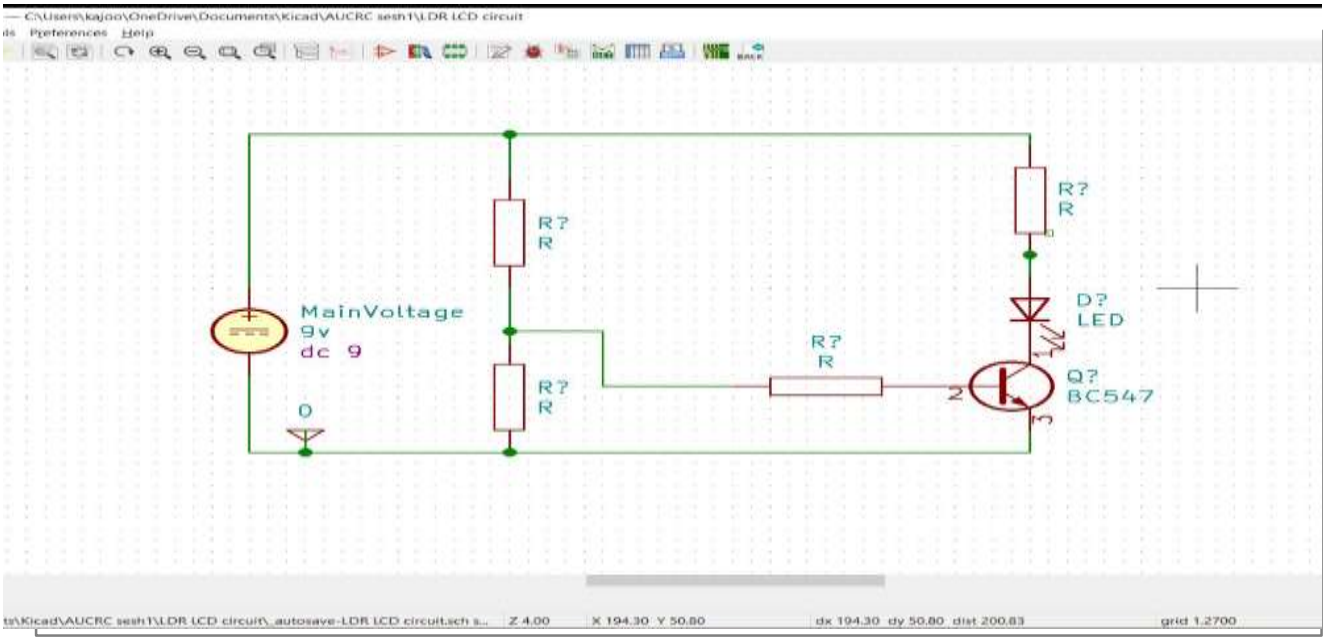


Fig 1. Schematic from KiCad

Reference circuit diagram (for understanding)

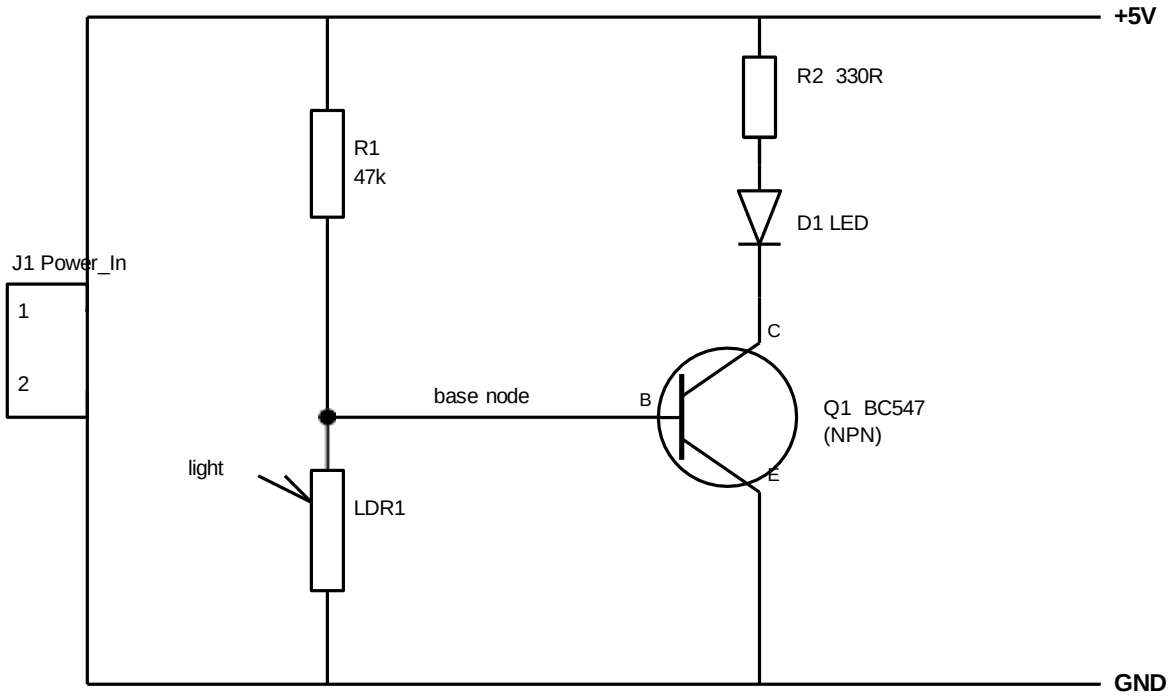
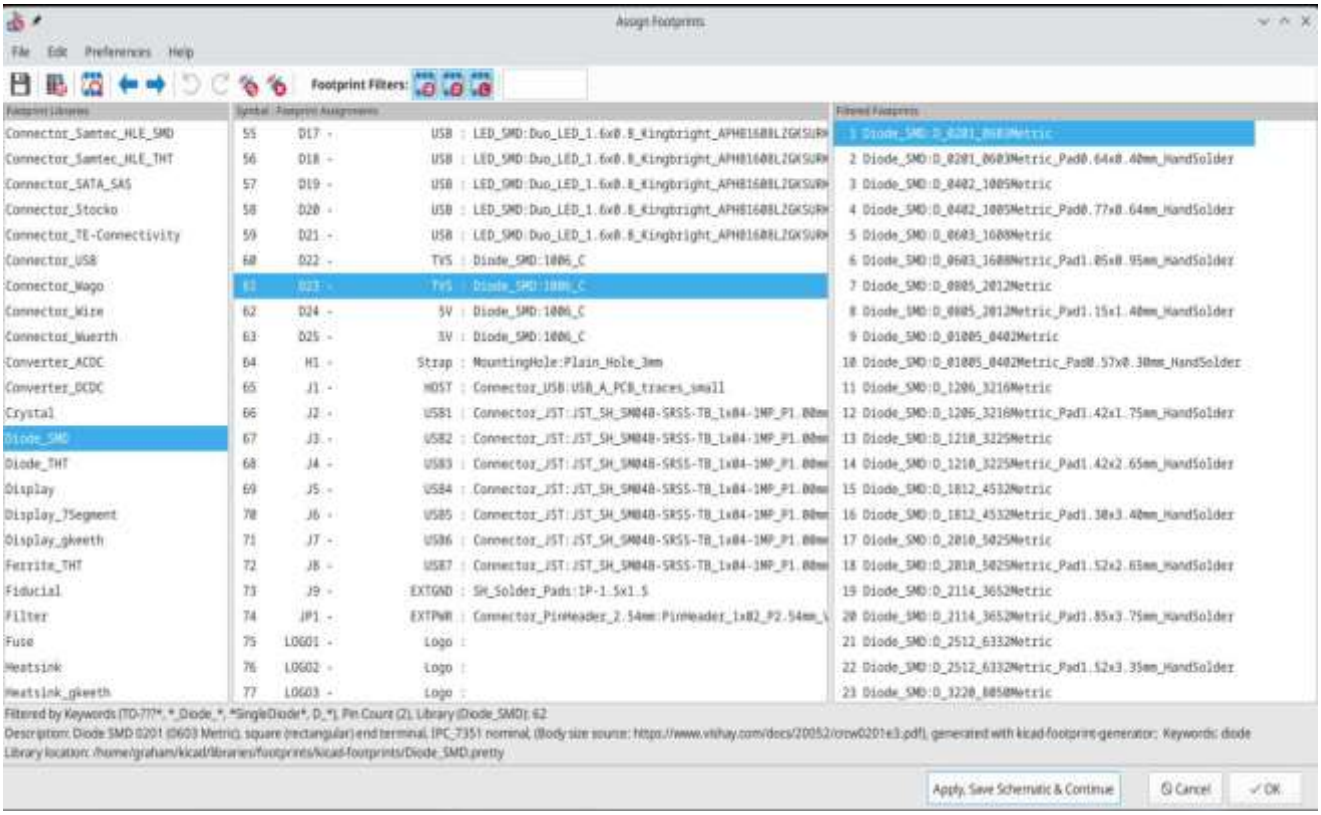


Fig 2. Simplified reference diagram of the circuit

3. Footprint Assignment

Ref	Component	Symbol	Footprint
R1, R2	Resistor	Device:R	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
LDR1	Light dependent resistor	Device:R_Photo	OptoDevice:R_LDR_5.1x4.3mm_P3.4mm_Vertical
Q1	BC547 NPN transistor	Transistor_BJT:BC547	Package_TO_SOT_THT:TO-92_Inline
D1	LED 5 mm	Device:LED	LED_THT:LED_D5.0mm
J1	Power connector	Connector:Conn_01x02_Pin	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical



4. PCB Layout After Routing

Board size approx. 40 x 30 mm. Track width 0.4 mm (signals) and 0.6 mm (power), clearance 0.25 mm, GND copper fill on the bottom layer. DRC result: 0 errors, 0 unconnected items.

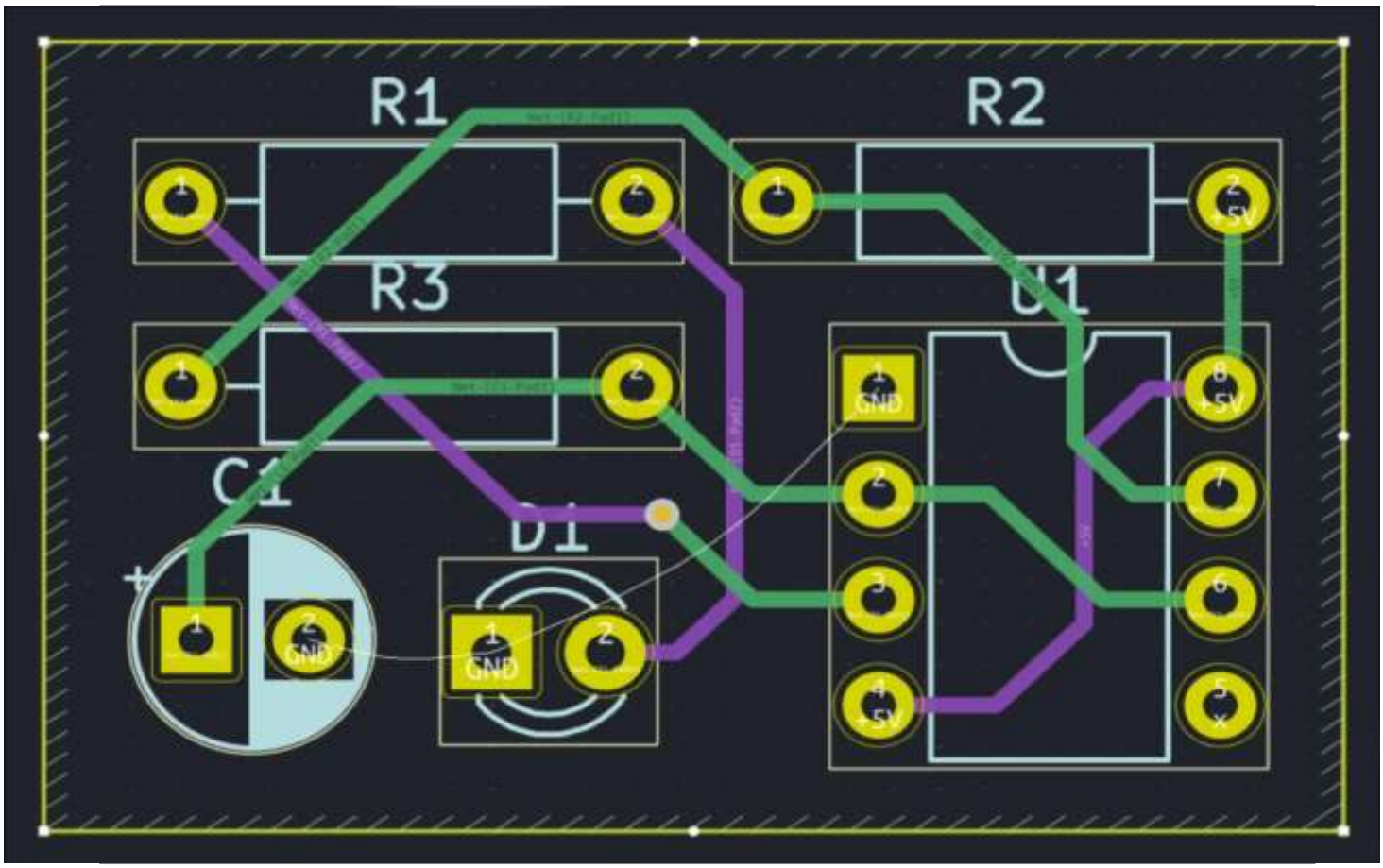
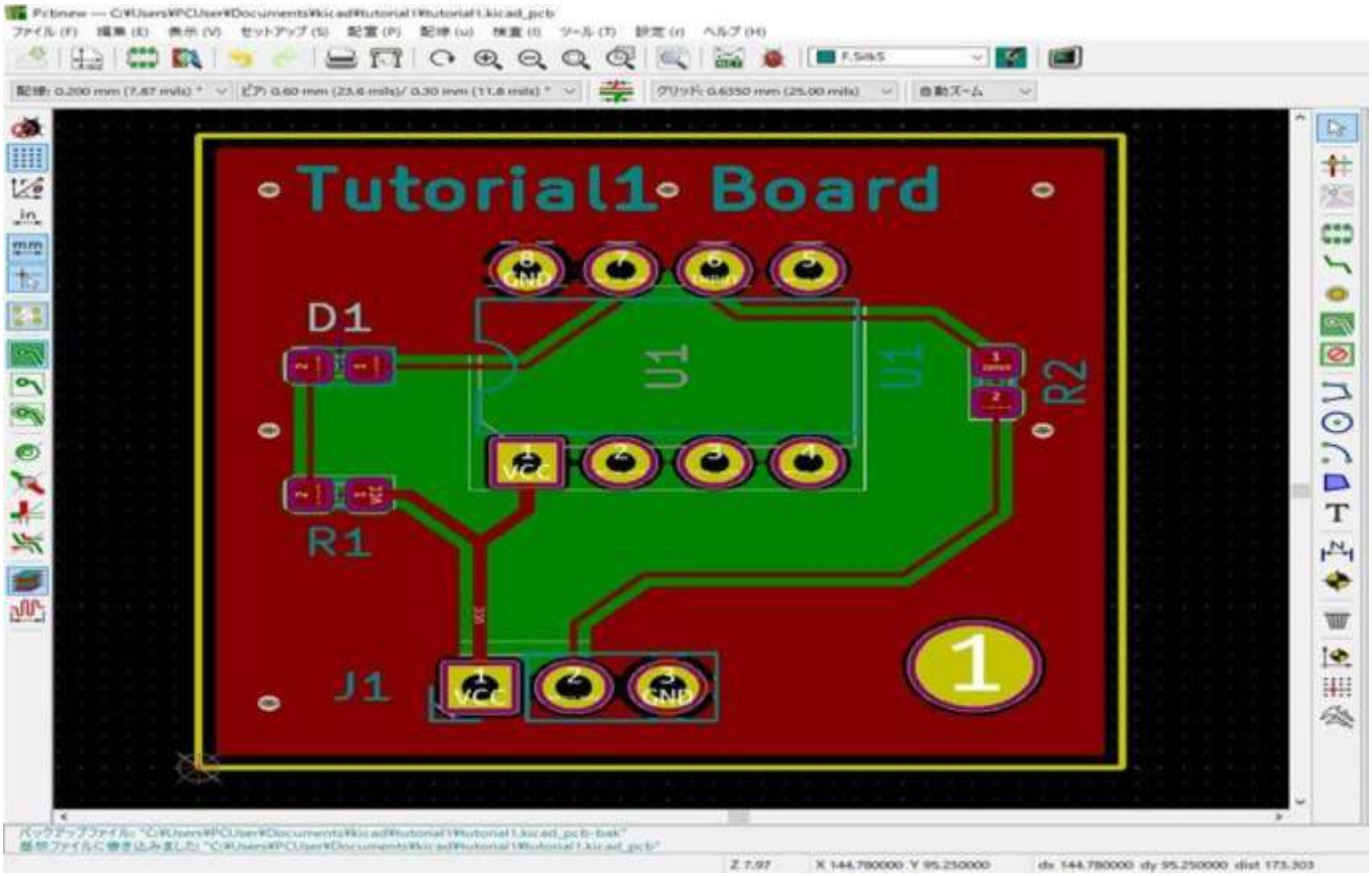


Fig 4. Footprints placed and routed

5. 2D PCB View



6. 3D PCB View

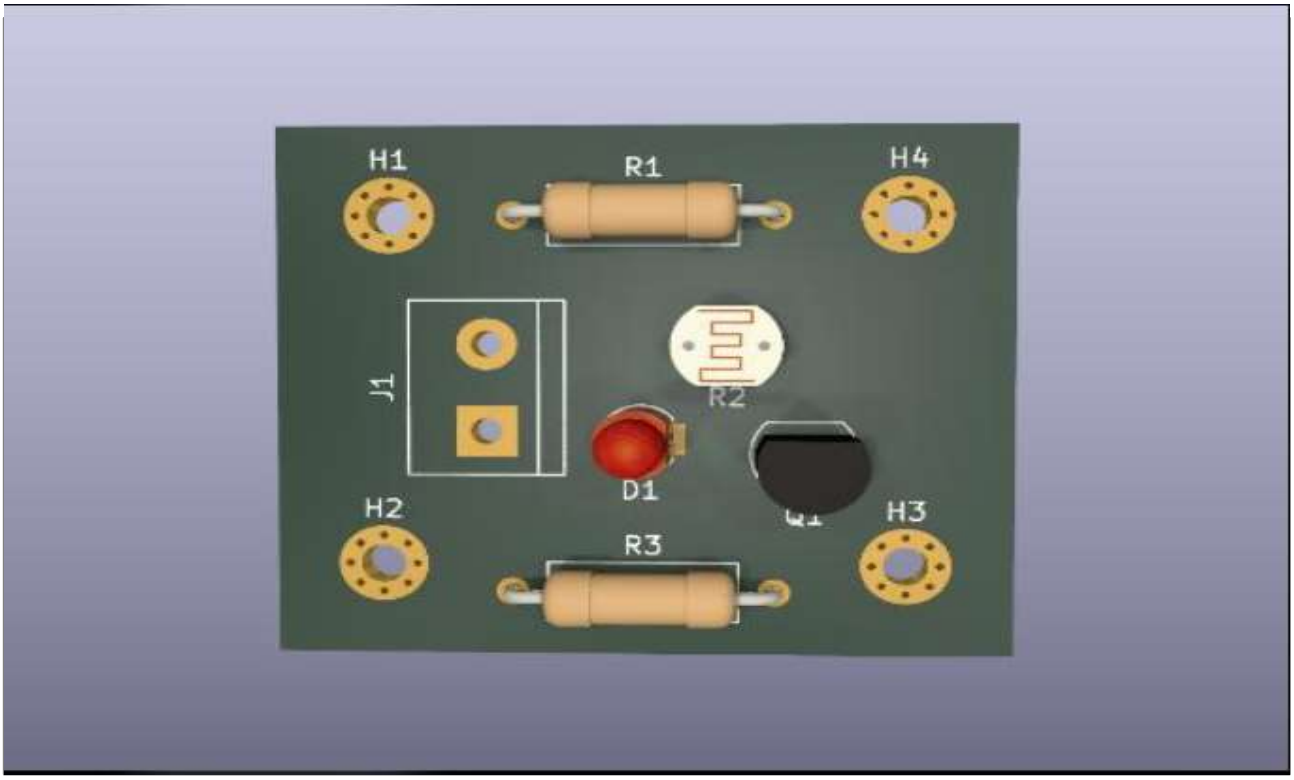


Fig 6. 3D view of the PCB

7. Bill of Materials (BOM)

Item	Reference	Value	Footprint	Qty
1	R1	47k	R_Axial_DIN0207 (THT)	1
2	R2	330R	R_Axial_DIN0207 (THT)	1
3	LDR1	LDR (5 mm)	R_LDR_5.1x4.3mm_P3.4mm	1
4	Q1	BC547	TO-92_Inline	1
5	D1	LED	LED_D5.0mm	1
6	J1	Power_In (2-pin)	PinHeader_1x02_P2.54mm	1

Gerber and drill files are uploaded to the GitHub repository together with this report.